WIC Redemptions, Household Food Security, and the Quality of Food Acquisitions

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Abstract

We estimate associations among the Special Supplemental Program for Women, Infants, and Children (WIC), households' diet quality, and food security among those who redeemed their WIC benefits compared with those who did not, using the National Household Food Acquisition Purchase Survey (FoodAPS) and augmented inverse probability weighting (AIPW). We use the Healthy Eating Index (HEI) 2010 and a binary indicator to measure households' food acquisition quality and food security, respectively. First, we find a positive association between WIC and diet quality when benefits are redeemed. Second, among HEI components, improvements are concentrated in fruit, whole grains, dairy, protein foods, seafood, and plant proteins among WIC-redeemed participants compared with WIC-eligible nonparticipants. Unlike diet-quality results, the food insecurity estimates do not suggest that WIC reduces household food insecurity among WIC-eligible households. Our findings highlight the importance of maximizing WIC benefit redemptions rather than simply increasing enrollment.

1. Introduction

Access to nutritious food is vital for health and human well-being. The Supplemental Nutrition Program for Women, Infants, and Children (WIC), the third-largest food assistance program in the U.S., primarily serves pregnant, postpartum, or breastfeeding women; infants; and children up to age 5 (Tiehen, 2020). Over time, WIC has provided nutritious foods, nutrition education, breastfeeding support, and referrals to healthcare and social services for economically disadvantaged, nutritionally at-risk women, infants, and children (Hodges et al., 2024). It is based on the idea that medical and developmental problems can be prevented or reduced through early nutritional intervention during childhood. Accordingly, WIC served more than 6 million people each month during fiscal year 2022 (Toossi & Jones, 2023), and around 40% of infants and 22% of children up to 5 years of age participated in the program in the United States during fiscal year 2021 (Hodges et al., 2024). Despite this broad reach and nutritional focus, understanding whether it effectively improves household nutritional well-being remains an important policy issue.

The goal of this study is to examine whether WIC participation is associated with household nutritional well-being across two key dimensions: household diet quality and food security. More specifically, we examine whether these associations differ when participants are distinguished by benefit redemption status and compared with WIC-eligible nonparticipants. Understanding these relationships is important because WIC participation alone may not capture actual use of WIC food benefits, and associations with household diet quality and food security may therefore differ between households that redeem their benefits and those that do not. This distinction helps us determine whether observed associations are attributable to participation in WIC or to the actual use of WIC food benefits.

Over the past several decades, the associations among WIC, birth, maternal health, and child health outcomes have been documented in several studies. Reviews of this literature generally conclude that WIC is associated with improved outcomes (Devaney, 2007; Fox et al., 2004; Owen & Owen, 1997), while also emphasizing persistent methodological concerns related to nonrandom program participation (Besharov & Germanis, 2001)[1]. Nevertheless, selection bias is a concern because many studies compare WIC participants with eligible nonparticipants (see Figueroa et al., 2025; Herman et al., 2024; Li et al., 2022; Thoma et al., 2023, 2024). If WIC participants are more health-conscious or better connected to health services than eligible nonparticipants, estimated associations may be overstated. In contrast, if WIC participants are more disadvantaged than eligible nonparticipants, estimated associations may understate the program's benefits (Bitler & Currie, 2005).

Recent studies have addressed these concerns using program benefit redemptions and empirical strategies designed to mitigate selection bias, though mostly for birth outcomes. For example, some studies use maternal fixed effects to compare births to the same mother with and without WIC participation, finding associations with reductions in low birth weight and small-for-gestational-age births (e.g., Currie & Rajani, 2015; Sonchak, 2016). Others use instrumental variables, narrow eligibility comparisons, or variation in county-level WIC rollout to address nonrandom selection (see Figlio et al., 2009; Gai & Feng, 2012; Hoynes et al., 2011).

[1] Devaney (2007) synthesized 35 years of WIC research and found that WIC participation was associated with improved maternal and child health outcomes, noting persistent challenges in identifying causal effects. Fox et al. (2004) reviewed WIC literature on nutrition and health and found that, despite selection bias, most studies reported positive effects of WIC on birth outcomes, nutrition, and health. Owen and Owen (1997) reviewed early evaluations of the WIC program and reported associations with improved birth outcomes, child growth, and reduced anemia.

Nevertheless, some studies have documented the benefit redemption as a more direct measure of WIC program utilization than participation alone. Early evidence documents variation in benefit redemption and identifies barriers such as transaction costs, administrative burden, and stigma that may prevent participants from fully using their WIC benefits (Rosenberg et al., 2003; Woelfel et al., 2004). A more recent line of research shows that replacing paper vouchers with electronic benefit transfer (eWIC) increases benefit redemption by reducing these barriers (Hanks et al., 2019).

Despite these advances, there has been comparatively little attention to whether benefit redemption influences the associations between WIC participation and household nutritional well-being, given the role of selection bias. Overwhelmingly, recent studies that have examined household nutritional outcomes and birth outcomes report positive associations between WIC participation and diet quality, food acquisition, nutrient intake, birth weight, and obesity (Mande & Flaherty, 2022; Smith & Valizadeh, 2024; Spence et al., 2024), with mixed findings on food security (Cho, 2022; Harper et al., 2022; Hazzard et al., 2023; Insolera et al., 2022; Whaley et al., 2023). These results are noteworthy because WIC food packages can influence household food acquisitions only when benefits are redeemed. Once WIC foods enter the household, they may be shared among household members, making household-level dietary outcomes an appropriate outcome while also complicating attribution to the intended recipient. Moreover, households that choose to redeem benefits may differ systematically from those that do not, even among WIC participants. Because of the selection issue, despite the growing body of evidence linking WIC participation to better dietary outcomes, evidence distinguishing associations attributable to program participation from those attributable to actual benefit redemption, particularly with respect to household food security, is limited.

To address this question, we use the USDA's FoodAPS data, a nationally representative survey containing detailed information on household food acquisitions, WIC participation, benefit redemption, and food security. An advantage of FoodAPS is that it allows us to distinguish WIC participants who redeemed their benefits during an interview period (typically a week) from those who did not. Since the sample wasn't stratified by time of month or WIC benefit delivery date, the samples of households using or not using WIC benefits should be approximately random. To determine how WIC contributes to the household nutritional well-being, we use two complementary outcomes: the quality of household food acquisitions, measured by the Healthy Eating Index (HEI)-2010, and household food security.

Empirically, given the lack of a suitable instrumental variable that could be used to isolate the effects of WIC participation, we estimate these associations using augmented inverse probability weighting (AIPW), a doubly robust estimator that combines propensity score weighting with outcome regression to account for differences in observed household characteristics[2]. We compare overall WIC participants, WIC participants who redeemed benefits, and WIC-eligible nonparticipants to distinguish associations related to program participation from those associated with actual benefit use. As expected, we find positive associations between household diet quality and households that redeemed WIC benefits, whereas associations with household food security are less strong than those with dietary quality. These findings highlight the importance of benefit redemption when evaluating WIC's associations with household nutritional well-being.

[2] This empirical approach builds on Fang et al. (2019), who use the same FoodAPS data and propensity score matching (PSM) to examine associations between WIC participation, benefit redemption, and the nutritional quality of household food purchases. We complement their analysis by employing a doubly robust AIPW estimator and by additionally examining household food security.

The paper is organized as follows. The next section describes the FoodAPS data, sample construction procedures, outcome measures, and the augmented inverse probability weighting (AIPW) framework used to estimate the associations between WIC participation, benefit redemption, household diet quality, and food security. We then present the main empirical results, beginning with determinants of WIC participation and covariate balance diagnostics, followed by the estimated associations between WIC participation, benefit redemption, household diet quality, and food security. Finally, we discuss the implications of our findings for WIC policy, acknowledge the study's limitations, and conclude.

2. Data and Methods

We use FoodAPS data, a nationally representative survey conducted between April 2012 and January 2013 and jointly developed by the U.S. Department of Agriculture's Economic Research Service (ERS) and Food and Nutrition Service (FNS), to provide comprehensive information on household food acquisition behavior in the United States (USDA, 2026). Of the 4,826 households, our final analysis sample consists of 935 households (see the detailed sample construction in Appendix Table A1 and Figure A1). Among these households, 507 are WIC-eligible nonparticipants, and 428 are WIC participants. Of the households participating, 146 redeemed their WIC benefits on at least one purchase occasion during the survey week, and 282 did not. It is important to note that FoodAPS records household food acquisitions only during the survey week, so redemption behavior is observed only within that interview period. Accordingly, households classified as nonredeemers are those for whom no WIC benefits were observed during the survey week; thus, this cannot be taken as evidence that they failed to redeem benefits throughout the entire benefit cycle.

FoodAPS contains detailed information on household demographic, socioeconomic, and food access characteristics that may be associated with both WIC participation and household nutritional well-being. As reported in Table 1, these variables include household education, monthly income, marital status, rural residence, race and ethnicity, the presence of WIC-eligible women, the number of young children and infants, nutrition knowledge, food access, vehicle ownership, and household size[3]. Most of these measures are constructed by aggregating individual-level information to the household level. Household diet quality is measured using the Healthy Eating Index (HEI)-2010, while household food insecurity is measured using the USDA FoodAPS food security module. Detailed variable definitions are provided in Appendix Table A2.

Measuring Household Diet Quality and Food Security

We measure household diet quality using the 2010 Healthy Eating Index (HEI-2010), a widely used measure of compliance with the 2010 Dietary Guidelines for Americans (USDA, 2010). The HEI, which was originally created in 1995, has been revised over time to reflect updates to the Dietary Guidelines for Americans (see Guenther et al., 2013, 2014). Because FoodAPS was conducted during 2012–2013, when the 2010 Dietary Guidelines were in effect, we use the HEI-2010 rather than the HEI-2015[4]. The HEI-2010 consists of 12 dietary components, including adequacy and moderation components, and is scored from 0 to 100, with higher scores indicating healthier food acquisitions[5]. Following Fang et al. (2019), we calculate the HEI-2010 using

[3] Highest household education refers to the highest educational attainment of any household member. The number of children younger than 5 years and infants is based on children aged 1-4 years and those younger than 1 year, respectively. A WIC-eligible woman indicator is set to 1 if the household includes a female member aged 14-49. Nutrition knowledge is measured using the FoodAPS nutrition knowledge index, and primary food access indicates that the household's primary food store is located within 10 miles of the residence.
[4] We use the HEI-2010 rather than the HEI-2015 because FoodAPS was collected during 2012–2013, when the 2010 Dietary Guidelines for Americans were current and the HEI-2015 had not yet been released.
[5] The HEI-2010 includes nine adequacy components (total fruit, whole fruit, total vegetables, greens and beans, whole grains, dairy, total protein foods, seafood and plant proteins, and fatty acids) and three moderation components (refined

household food acquisitions rather than individual dietary intake, which provides a measure of the nutritional quality of foods acquired by the household during the survey week.

Table 1. Descriptive statistics by WIC participation and redemption status

	All	Non-WIC	WIC	Redeemed	Not redeemed
Panel A: Diet quality					
HEI-2010 score	49.050	50.165	46.301	55.012	41.900
Panel B: Food security					
Food insecure	0.197	0.151	0.311	0.266	0.333
Panel C: Household demographics					
Rural household (%)	26.026	26.272	25.420	26.188	25.032
Married household (%)	64.036	69.736	49.976	54.128	47.878
Non-Hispanic White (%)	59.863	65.373	46.269	59.011	39.831
Non-Hispanic Black (%)	17.348	16.090	20.451	13.858	23.783
Hispanic (%)	30.068	23.945	45.173	39.836	47.869
Highest household education	3.085	3.290	2.582	2.647	2.549
Panel D: Economic characteristics					
Monthly household income ($1,000s)	5.088	5.920	3.035	3.034	3.036
Panel E: WIC eligibility and household composition					
Number of children age <=5	1.330	1.312	1.376	1.321	1.404
Number of infants age <=1	0.418	0.355	0.571	0.606	0.553
WIC-eligible woman present (%)	97.358	96.465	99.561	100.000	99.340
Panel F: Nutrition knowledge and food access					
Nutrition knowledge (%)	91.996	93.079	89.324	91.543	88.203
Primary store within 10 miles (%)	90.735	91.736	88.264	89.162	87.810
Household has vehicle (%)	90.050	93.662	81.141	82.472	80.469
Panel G: Household size					
Household size	4.181	4.039	4.530	4.518	4.537

Notes: Means are weighted using the FoodAPS household survey weight. Food insecure is a binary indicator coded as 1 if the household is food insecure and 0 otherwise. Demographic and access indicators are reported as percentages. The analytic sample is restricted to WIC-categorically eligible households with valid food-at-home nutrition records and complete covariate information. WIC redemption is measured during the FoodAPS food-at-home event reporting period. Source: USDA FoodAPS, 2012-2013.

grains, sodium, and empty calories). Higher scores indicate greater consumption of adequacy components and lower consumption of moderation components (Guenther et al., 2013; USDA, 2010).

Each food item recorded in FoodAPS is linked to the USDA's Food Pattern Equivalents Database (FPED) through the Food and Nutrient Database for Dietary Studies (FNDDS) or the National Nutrient Database for Standard Reference (SR) to obtain the quantities required for HEI-2010 calculation (Mancino et al., 2018). USDA Economic Research Service researchers developed the nutrient coding procedures and associated computer programs used to aggregate food items into HEI components at the household level (USDA, 2016). Following this approach, we calculate each household's HEI-2010 score using all food acquisitions reported during the survey week. Thus, higher HEI-2010 scores indicate higher overall nutritional quality of household food acquisitions. Nevertheless, for adequacy components, higher scores indicate higher intakes; for moderation components, higher scores indicate lower intakes.

We measure household food security using the 10-item, 30-day Adult Food Security Survey Module (AFSSM) administered during the final FoodAPS interview after completion of the one-week food acquisition diary. The module consists of ten questions assessing progressively more severe experiences of food-related hardship over the previous 30 days, ranging from concerns about food running out to going an entire day without eating because of insufficient resources (Gregory et al., 2019). Responses are coded following the USDA's standard scoring procedures, whereby households with three or more affirmative responses are classified as food insecure (USDA, 2012)[6]. Accordingly, we construct a binary indicator that equals 1 if the household is food insecure and 0 otherwise.

[6] Affirmative responses are defined according to USDA guidelines. For the first three questions, responses of "often" or "sometimes" are counted as affirmative. For the frequency questions regarding meal reductions and not eating for an entire day, responses are considered affirmative if the condition occurred on three or more days during the previous 30 days. All remaining items are coded affirmatively for a "yes" response. Households with three or more affirmative responses are classified as food insecure.

Characteristics of WIC, Redemption, and WIC-Eligible Nonparticipant Households

As shown in Table 1, the average HEI-2010 score is 50.17 for WIC-eligible nonparticipants and 46.30 for WIC participants. The last two columns of the table further distinguish WIC participants by redemption status. Here, the difference is substantial: households that redeemed WIC benefits have an average HEI-2010 score of 55.01 compared with 41.90 among those that did not redeem benefits. The last two columns of Table 1 also reveal meaningful differences in household food security. The prevalence of food insecurity is 26.6% among households that redeemed WIC benefits compared with 33.3% among those that did not. Both groups, however, exhibit higher rates of food insecurity than WIC-eligible nonparticipants (15.1%). Consistent with this pattern, WIC households have lower monthly income, lower household education, lower marriage rates, and lower vehicle access than eligible nonparticipants. Nevertheless, given these differences across WIC participation and redemption groups, the descriptive patterns alone cannot indicate whether WIC participation or benefit redemption is associated with household diet quality and food security after accounting for observed household characteristics.

Selection into WIC Participation and Benefit Redemption

As discussed above, estimating the associations between WIC participation and household nutritional well-being is challenging because participation in WIC is not randomly assigned. Compared with WIC-eligible nonparticipants, participating households may differ in demographic, socioeconomic, and nutrition-related characteristics that are also associated with household diet quality and food security. Likewise, among participating households, benefit redemption is unlikely to occur randomly, as households that redeem their benefits may differ from those that do not in ways related to shopping behavior, access to food retailers, nutrition awareness, or other observed characteristics. As a result, to examine whether WIC participation and benefit redemption are associated with household diet quality and food security, we must first account for systematic differences in observed household characteristics across the comparison groups.

Empirical Approach

Building on the descriptive comparisons presented above, we next estimate adjusted associations using augmented inverse probability weighting (AIPW). Similar to other propensity score methods, AIPW, which was developed by Robins et al., (1995), seeks to reduce bias arising from observable differences between treated and comparison households by balancing measured baseline characteristics across groups (DiPrete & Gangl, 2004; Rosenbaum, 2002; Rosenbaum & Rubin, 1983). Unlike propensity score matching alone, however, AIPW combines inverse probability weighting with outcome regression, providing doubly robust estimates when either model is correctly specified (Funk et al., 2011; Robins et al., 1994; Wooldridge, 2010).

Following the potential outcomes framework of Rosenbaum and Rubin (1983), let T_i denote the treatment indicator for household i, where $T_i = 1$ if the household belongs to the treatment group and $T_i = 0$ otherwise. Because our analysis considers three comparisons, we estimate separate models for (i) all WIC participants versus WIC-eligible nonparticipants, (ii) WIC participants who redeemed benefits versus WIC-eligible nonparticipants, and (iii) WIC participants who did not redeem benefits versus WIC-eligible nonparticipants. For each comparison, let $Y_i(1)$ and $Y_i(0)$ denote the potential outcomes under treatment and control, respectively, where Y_i represents either household diet quality or household food security. The observed outcome is

$$Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0), \tag{1}$$

where only one potential outcome is observed for each household. Therefore, the individual treatment effect cannot be observed directly. The parameter of interest is the average treatment effect (ATE),

$$ATE = E[Y_i(1) - Y_i(0)], \tag{2}$$

which measures the average difference in household nutritional well-being associated with WIC participation or benefit redemption.

Because WIC participation is not randomly assigned, the counterfactual outcomes required to estimate the average treatment effect (ATE) are not directly observed. In particular, the potential outcome under nonparticipation for WIC participants, $E[Y(0) \mid T = 1]$, is unobservable. Replacing this counterfactual with the observed outcomes of eligible nonparticipants, $E[Y(0) \mid T = 0]$, would generally produce biased estimates because households that participate in WIC may differ systematically from eligible nonparticipants in ways that also affect the outcomes of interest. Therefore, identification of the ATE using observational data requires two standard assumptions (e.g., Drichoutis et al. (2009)). First, conditional independence (also called "exogeneity," ignorability," "unconfoundedness," or "selection on observables") assumes that treatment assignment is independent of the potential outcomes after conditioning on observed household characteristics X_i (Angrist et al., 1996),

$$\{Y_i(1), Y_i(0)\} \perp T_i \mid X_i, \tag{3}$$

where X_i includes household demographic, socioeconomic, nutrition, and food access characteristics reported in Table 1. This assumption implies that the missing counterfactual outcome for each treated household can be recovered from otherwise similar comparison households (Hernan, 2004). In this setting, conditioning on observed characteristics removes systematic differences between treatment groups, so that observed outcome differences are attributable to treatment rather than underlying household characteristics. Second, the overlap assumption requires every household to have a positive probability of belonging to either treatment group,

$$0 < P(T_i = 1 \mid X_i) < 1, \tag{4}$$

Following Rosenbaum and Rubin (1983) and Caliendo and Kopeinig (2008), these assumptions imply that households with similar observed characteristics are comparable across treatment groups. The propensity score is therefore defined as

$$e(X_i) = P(T_i = 1 \mid X_i), \tag{5}$$

In this study, to estimate the ATE, we employ augmented inverse probability weighting (AIPW), which combines inverse probability weighting based on the estimated propensity score with regression adjustment for the outcome. Specifically, the propensity score $e(X_i)$ is estimated using a logit model, while separate outcome models are estimated for treated and comparison households. The AIPW estimator is given by

$$\hat{C}_{AIPW}(X_i) = \left(\underbrace{\frac{T_i Y_i}{\hat{e}(X_i)}}_{IPW} - \underbrace{\frac{T_i - \hat{e}(X_i)}{\hat{e}(X_i)} \widehat{m}_1(X_i)}_{Augmentation} \right) - \left(\underbrace{\frac{(1 - T_i) Y_i}{1 - \hat{e}(X_i)}}_{IPW} + \underbrace{\frac{T_i - \hat{e}(X_i)}{1 - \hat{e}(X_i)} \widehat{m}_0(X_i)}_{Augmentation} \right), \tag{6}$$

followed by

$$\widehat{ATE}_{AIPW} = \frac{1}{N}\sum_{i=1}^{N} \hat{C}_{AIPW}\,(X_i), \tag{7}$$

where T_i is the treatment indicator, Y_i is the observed outcome (HEI-2010 score or food insecurity), $\hat{e}(X_i)$ is the estimated propensity score, and $\widehat{m}_1(X_i)$ and $\widehat{m}_0(X_i)$ denote the predicted outcomes under treatment and control, respectively. In equation (6), the first term in each bracket represents the inverse probability weighting (IPW) component, whereas the second term is the augmentation component (see Kurz (2022) for detailed proofs). By combining these two components, the AIPW estimator is doubly robust because it yields consistent estimates of the ATE provided that either the propensity score model or the outcome regression model is correctly specified.

3. Results and Discussion

Before estimating the AIPW models, we first examine baseline differences between WIC participants and WIC-eligible nonparticipants. As reported in Appendix Table A4, the two groups differ across a number of demographic, socioeconomic, and household characteristics, suggesting that WIC participation is not randomly distributed in the sample. In particular, participating households exhibit lower average household diet quality, higher rates of food insecurity, lower educational attainment and income, and a greater presence of WIC-eligible women and infants than eligible nonparticipants[7].

As shown in Table 4, consistent with previous studies of WIC participation, households with higher educational attainment and higher monthly income are significantly less likely to participate in

[7] Appendix Table A4 reports unadjusted differences and standardized differences between WIC participants and WIC-eligible nonparticipants. Several covariates exhibit standardized differences exceeding commonly used balance thresholds (Austin, 2009), indicating meaningful baseline differences between the two groups and motivating adjustment using the AIPW framework.

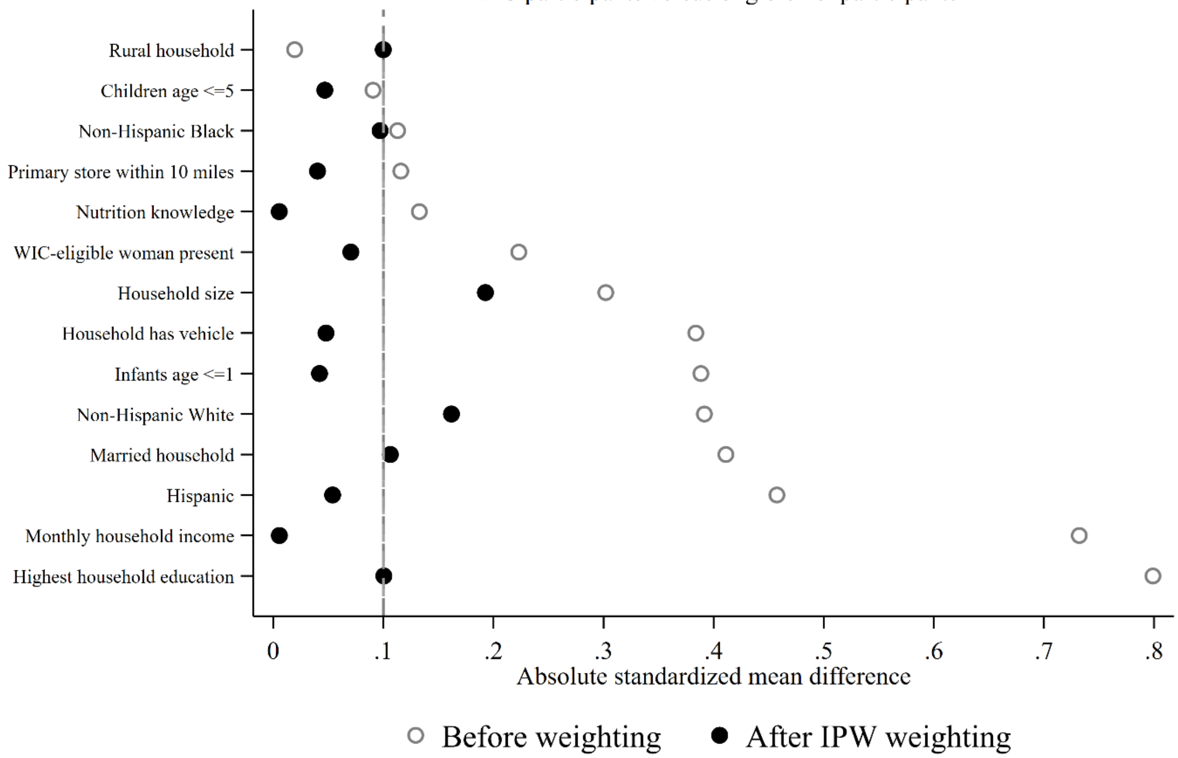
WIC (Anderson & Whaley, 2025; Guan et al., 2023), whereas Hispanic households, households with infants, and households containing a WIC-eligible woman are more likely to participate (Collin et al., 2023; Pulvera et al., 2022; Zhang et al., 2022). In contrast, households with more children aged one to five are less likely to participate, consistent with evidence that WIC participation declines as children grow older (M. P. Bitler et al., 2003; Tiehen & Jacknowitz, 2008). Household size is positively associated with participation (Anderson & Whaley, 2025; Collin et al., 2023), while rural residence, marital status, nutrition knowledge, proximity to the primary food store, and vehicle ownership are not statistically significant predictors. The estimated propensity scores from this model are subsequently used to construct the AIPW estimator.

Assessing Covariate Balance

As discussed in the previous section, AIPW relies on balancing observed household characteristics between treatment and comparison groups to reduce bias arising from nonrandom WIC participation. To assess whether inverse probability weighting improves covariate balance, we compare absolute standardized differences before and after weighting for all covariates included in the propensity score model. Following Rubin (1991) and Austin (2009), absolute standardized differences provide a scale-independent measure of covariate balance that is not affected by sample size. As shown in Figure 1, before weighting, several covariates exhibit imbalance between treatment groups. In particular, household education, monthly household income, Hispanic ethnicity, marital status, vehicle ownership, the presence of infants, and the presence of a WIC-eligible woman all display standardized differences well above the commonly used 0.10 threshold (Austin, 2009). After applying inverse probability weighting, the magnitude of these differences declines markedly across nearly all covariates. Most standardized differences are below 0.10, while the remaining covariates are more balanced than in the unweighted sample.

Figure 1. Covariate Balance Before and After Weighting

WIC participants versus eligible nonparticipants



Benefit Redemptions, Diet Quality, and Food Security

Table 2 reports AIPW estimates for the associations between WIC participation, benefit redemption, and the quality of household food acquisitions. Panel A compares all WIC participants with WIC-eligible nonparticipants, regardless of whether benefits were redeemed during the survey week. We estimate a positive and statistically significant average treatment effect of 2.413 HEI-2010 points. Given that the average HEI-2010 score among WIC participants is 46.30 (see Table 1), this estimate represents an improvement of approximately 5.2 percent in the nutritional quality of household food acquisitions. This finding is consistent with previous studies reporting positive associations between WIC participation and dietary quality, food purchases, and nutrient adequacy (e.g., Black et al., 2012; Fang et al., 2019; Fox et al., 2004; Mande & Flaherty, 2022; Smith & Valizadeh, 2024).

Table 2. WIC Participation, Redemption, and HEI-2010

Treatment Comparison	ATE
Panel A: All WIC Participants vs Eligible Nonparticipants	2.413**
	(1.071)
Observations	935
Treated observations	428
Control observations	507
Overlap violators dropped	0
Panel B: Redeemed WIC vs Eligible Nonparticipants	11.848***
	(1.518)
Observations	637
Treated observations	146
Control observations	491
Overlap violators dropped	16
Panel C: Did Not Redeem WIC vs Eligible Nonparticipants	-1.839
	(1.438)
Observations	789
Treated observations	282
Control observations	507
Overlap violators dropped	0

Notes: The table reports augmented inverse probability weighting (AIPW) estimates of the average treatment effect (ATE) on the HEI-2010 score. Standard errors are reported in parentheses. Estimates are weighted using the FoodAPS household survey weight. Covariates include rural residence, marital status, race and ethnicity, household education, income, number of children age five or younger, number of infants age one or younger, presence of a WIC-eligible woman, nutrition knowledge, store access, vehicle ownership, and household size. Overlap violators are observations identified by Stata's osample option as violating the overlap assumption and are excluded from the subgroup AIPW models when necessary. Asterisks indicate statistical significance: * $p<0.10$, ** $p<0.05$, *** $p<0.01$. Source: USDA FoodAPS, 2012-2013.

As shown in Panel B of Table 2, for households that redeemed their WIC benefits, we estimate a positive and highly statistically significant average treatment effect of 11.848 HEI-2010 points relative to WIC-eligible nonparticipants. Given that the average HEI-2010 score among redeemed households is 55.01 (see Table 1), this estimate corresponds to an improvement of approximately 21.5 percent in the nutritional quality of household food acquisitions. In contrast, in panel C, households that participated in WIC but did not redeem their benefits exhibit no statistically significant difference in HEI-2010 scores relative to eligible nonparticipants, with an estimated effect of −1.839 points. These findings suggest that the positive association between WIC participation and household diet quality is concentrated among households that actually redeem their WIC food benefits rather than among all enrolled participants. This pattern is consistent with the mechanism underlying the WIC program, as food packages can improve the nutritional quality of household food acquisitions only when benefits are redeemed. More broadly, the results extend the findings of Fang et al. (2019) by showing that the strong association between benefit redemption and household diet quality remains when estimated using a doubly robust AIPW framework.

Table 3 reports the AIPW estimates for the individual HEI-2010 components among households that redeemed WIC benefits. Beginning with the adequacy components, redeemed households exhibit significant improvements in total fruit, whole fruit, greens and beans, whole grains, dairy, total protein foods, seafood and plant proteins, and fatty acids, whereas no significant difference is observed for total vegetables. This pattern is broadly consistent with Fang et al. (2019) and Smith and Valizadeh (2024), both of which report that WIC-related improvements in dietary quality are concentrated in selected HEI components, particularly fruits, whole grains, dairy, protein foods, and reductions in less healthy dietary components.

Table 3. AIPW Estimates for HEI Components: Redeemed WIC

HEI Component	ATE
Adequacy Components	
Total fruits	1.111***
	(0.239)
Whole fruits	0.658**
	(0.271)
Total vegetables	0.200
	(0.204)
Greens and beans	0.348*
	(0.201)
Whole grains	1.146**
	(0.477)

Dairy	1.118*
	(0.679)
Total protein	0.700***
	(0.258)
Seafood and plant protein	0.863***
	(0.312)
Fatty acids	0.906**
	(0.435)
Moderation Components	
Refined grains	-0.395
	(0.922)
Sodium	0.940*
	(0.501)
Empty calories	4.252***
	(1.487)
Observations	637
Redeemed WIC households	146
Eligible nonparticipants	491
Overlap violators dropped	16

Notes: The table reports augmented inverse probability weighting (AIPW) estimates of the average treatment effect (ATE) of redeemed WIC participation on HEI-2010 component scores. The comparison group is WIC-categorically eligible nonparticipants. Standard errors are reported in parentheses. Estimates are weighted using the FoodAPS household survey weight. Covariates include rural residence, marital status, race and ethnicity, household education, income, number of children age five or younger, number of infants age one or younger, presence of a WIC-eligible woman, nutrition knowledge, store access, vehicle ownership, and household size. To satisfy the overlap assumption, 16 observations identified as overlap violators by the treatment model were excluded prior to estimation. Higher scores indicate better dietary quality for all HEI-2010 components, except for moderation components. Asterisks indicate statistical significance: * $p<0.10$, ** $p<0.05$, *** $p<0.01$. Source: USDA FoodAPS, 2012-2013.

Table 4 reports the AIPW estimates for household food insecurity. In contrast to the diet-quality results, we find no evidence that WIC participation or benefit redemption is associated with lower household food insecurity in this sample. The estimate for all WIC participants is positive and only weakly significant, indicating a 5.9 percentage point higher probability of food insecurity relative to eligible nonparticipants, while the estimates for redeemed WIC and nonredeemed WIC are not statistically significant. This finding differs from studies showing that losing WIC increases food hardship, including Cho (2022), who finds that aging out of WIC increases child food insecurity by at least 1.1 percentage points, and M. Bitler et al. (2023), who find that adult women experience higher food insecurity when children lose WIC eligibility. It also contrasts with the broader review evidence that WIC generally shields households from food insecurity (Smith & Gregory, 2023). One possible explanation is that our outcome is household food insecurity, while WIC benefits are supplemental and targeted to specific foods; therefore, redemption may improve the nutritional quality of household food acquisitions without being large enough to offset the broader economic constraints that generate food insecurity. This interpretation is consistent with Smith and Gregory's (2023) framework that food insecurity reflects both quantity and quality dimensions of food hardship, which need not move together.

Table 4. AIPW Estimates: Food Insecurity

Treatment Comparison	ATE
All WIC Participants vs Eligible Nonparticipants	0.059*
	(0.032)
Observations	935
Treated observations	428
Control observations	507
Overlap violators dropped	0
Redeemed WIC vs Eligible Nonparticipants	0.062
	(0.045)
Observations	637
Treated observations	146
Control observations	491
Overlap violators dropped	16
Did Not Redeem WIC vs Eligible Nonparticipants	0.054
	(0.038)
Observations	789
Treated observations	282
Control observations	507
Overlap violators dropped	0

Notes: The table reports augmented inverse probability weighting (AIPW) estimates of the average treatment effect (ATE) on household food insecurity. Food insecurity is a binary indicator coded as 1 if the household is food insecure and 0 otherwise. Standard errors are reported in parentheses. Estimates are weighted using the FoodAPS household survey weight. Covariates include rural residence, marital status, race and ethnicity, household education, income, number of children age five or younger, number of infants age one or younger, presence of a WIC-eligible woman, nutrition knowledge, store access, vehicle ownership, and household size. Overlap violators are observations identified by Stata's osample option as violating the overlap assumption and are excluded from subgroup AIPW models when necessary. Asterisks indicate statistical significance: * p<0.10, ** p<0.05, *** p<0.01. Source: USDA FoodAPS, 2012-2013.

Robustness Check Using Machine Learning

The AIPW estimator we used in our main analysis relies on correctly specifying either the propensity score model or the outcome regression model. Although this doubly robust property provides protection against misspecification of one model, the functional form we used to estimate treatment probabilities may still be restrictive if the underlying relationship between WIC participation and observed household characteristics is highly nonlinear or involves complex interactions (Busso et al., 2014). To further assess the robustness of our findings, we apply an alternative nonparametric supervised machine learning approach based on generalized random forests (GRF), which imposes fewer assumptions on the underlying data-generating process (Athey, 2018; Athey et al., 2019; Wager & Athey, 2018). Generalized random forests use an "honest" sample-splitting procedure, where one subsample is used to build the trees and another is used to estimate treatment effects[8].

Generalized random forest estimates are reported in Table 5 and are broadly consistent with the main AIPW results discussed earlier. Specifically, we find similar estimates for all WIC participants (ATT = 2.510), households that redeemed WIC benefits (ATT = 11.537), and households that did not redeem benefits (ATT = −1.860). These results reinforce our conclusion that the positive association between WIC participation and household diet quality is driven by households that actually redeemed WIC benefits during the survey week. The component-level estimates also follow the same pattern as the main results, with positive and statistically significant improvements in total fruit, whole fruit, greens and beans, whole grains, dairy, total protein foods, seafood and plant proteins, fatty acids, sodium, and empty calories. For food insecurity, the generalized random forest results continue to provide no evidence that redeemed WIC participation reduces household food insecurity, which is consistent with the AIPW results and suggests that WIC redemptions are more strongly associated with diet quality than with broader household food security.

Table 5. Robustness check using Machine Learning: Generalized Random Forest

Outcome / treatment comparison	**ATT estimate**
All WIC participants	2.510**
	(1.058)
Subsample that redeemed WIC	11.537***
	(1.384)
Subsample that did not redeem WIC	-1.860
	(1.165)
Food insecurity, all WIC participants	0.078***
	(0.030)
Food insecurity, redeemed WIC	0.064
	(0.043)
Food insecurity, did not redeem WIC	0.099***
	(0.034)
HEI component, subsample that redeemed WIC	
Adequacy components	
Total fruits	0.924***

[8] We implement the estimator using the grf package in R developed by Athey et al. (2019).

	(0.157)
Whole fruits	0.586***
	(0.194)
Total vegetables	0.096
	(0.164)
Greens and beans	0.368**
	(0.184)
Whole grains	1.522***
	(0.290)
Dairy	1.525***
	(0.346)
Total protein	0.662***
	(0.178)
Seafood and plant protein	0.969***
	(0.217)
Fatty acids	0.628*
	(0.371)
Moderation components	
Refined grains	0.263
	(0.340)
Sodium	0.600*
	(0.330)
Empty calories	3.275***
	(0.639)

Notes: The table reports generalized random forest estimates of the average treatment effect of WIC participation on HEI-2010, household food insecurity, and HEI component scores. Standard errors are reported in parentheses. The models include the same covariates used in the main AIPW specifications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: USDA FoodAPS, 2012-2013.

4. Conclusion

This study examines whether WIC participation and benefit redemption are associated with household diet quality and food security among WIC-eligible households. Using FoodAPS and AIPW, we find that WIC participation is positively associated with household diet quality, but this association is concentrated among households that redeemed WIC benefits during the survey week. Households that redeemed benefits had substantially higher HEI-2010 scores than eligible nonparticipants, while households that participated but did not redeem benefits showed no statistically significant difference. Component-level results show that improvements are concentrated in fruits, whole grains, dairy, protein foods, seafood and plant proteins, and selected moderation components. In contrast, we find no evidence that WIC participation or redemption reduces household food insecurity in this sample. These results suggest that WIC food benefits are more directly connected to the nutritional quality of household food acquisitions than to broader household food security.

This study has several limitations. First, FoodAPS observes food acquisitions during only one survey week, so households classified as nonredeemers may have redeemed benefits at another point in the benefit cycle. Second, FoodAPS measures household food acquisitions rather than actual individual consumption, so we cannot determine whether WIC foods were consumed by the intended recipient or shared with other household members. Third, because FoodAPS is cross-sectional, the analysis cannot assess longer-term effects of WIC participation, nutrition education, or continued benefit use. Finally, the WIC sample is relatively small, which limits additional subgroup analysis. Even with these limitations, the findings highlight the importance of benefit redemption when evaluating WIC's role in improving household nutritional well-being.

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Appendix

Table A1. Sample construction

	Sample	Relevant Sample (%)	Full FoodAPS Sample (%)
Full FoodAPS household sample	4826	100.00	100.00
WIC-categorically eligible households	2914	60.38	60.38
WIC eligible with valid WIC participation status	1007	34.56	20.87
WIC eligible with valid FAH HEI records	995	98.81	20.62
Analytic file before complete-case restriction	995	100.00	20.62
Final complete-case analytic sample	935	93.97	19.37
Eligible nonparticipants	507	54.22	10.51
All WIC participants	428	45.78	8.87
Redeemed WIC	146	15.61	3.03

Did not redeem WIC	282	30.16	5.84

Notes: Column 2 reports the percent of the relevant denominator. For the first six rows, this is the percent retained from the previous sample-construction step. For the final four rows, this is the percent of the final complete-case analytic sample. Column 3 reports the percent of the full FoodAPS household sample. The full FoodAPS household sample is based on the USDA National Household Food Acquisition and Purchase Survey. WIC-categorically eligible households are households with at least one member categorically eligible for WIC. Valid FAH HEI records refer to households with valid food-at-home nutrition records used to construct the HEI-2010 score. The final analytic sample is restricted to WIC-categorically eligible households with valid WIC participation status, valid HEI outcomes, and complete covariate information. WIC redemption is measured during the FoodAPS food-at-home event reporting period. Source: USDA FoodAPS, 2012-2013.

Table A2. Variable Definitions and Data Sources

Variable	Definition	Coding / Measurement	Data Source
HEI-2010 Score	Household Healthy Eating Index score based on food-at-home acquisitions	Continuous (0–100); higher values indicate better diet quality	FoodAPS
Food Insecure	Household classified as food insecure	1 = food insecure; 0 = food secure	FoodAPS USDA Food Security Module
Very Low Food Security	Household classified as having very low food security	1 = very low food security; 0 = otherwise	FoodAPS USDA Food Security Module
Food Secure	Household classified as food secure	1 = food secure; 0 = food insecure	FoodAPS USDA Food Security Module
WIC Participant	Household participates in WIC	1 = participant; 0 = eligible nonparticipant	FoodAPS
Redeemed WIC	Participating household redeemed WIC benefits during FAH acquisition period	1 = redeemed; 0 = eligible nonparticipant	FoodAPS
Did Not Redeem WIC	Participating household did not redeem WIC benefits during FAH acquisition period	1 = participant without redemption; 0 = eligible nonparticipant	FoodAPS

Variable	Definition	Coding / Measurement	Data Source
Total Fruits	HEI component measuring total fruit consumption	0–5 points	HEI-2010 / FoodAPS
Whole Fruits	HEI component measuring whole fruit consumption	0–5 points	HEI-2010 / FoodAPS
Total Vegetables	HEI component measuring vegetable consumption	0–5 points	HEI-2010 / FoodAPS
Greens and Beans	HEI component measuring greens and legumes consumption	0–5 points	HEI-2010 / FoodAPS
Whole Grains	HEI component measuring whole grain consumption	0–10 points	HEI-2010 / FoodAPS
Dairy	HEI component measuring dairy consumption	0–10 points	HEI-2010 / FoodAPS
Total Protein Foods	HEI component measuring protein food consumption	0–5 points	HEI-2010 / FoodAPS
Seafood and Plant Proteins	HEI component measuring seafood and plant protein consumption	0–5 points	HEI-2010 / FoodAPS
Fatty Acids	HEI component measuring ratio of unsaturated to saturated fatty acids	0–10 points	HEI-2010 / FoodAPS
Refined Grains	HEI moderation component measuring refined grain intake	0–10 points; higher scores indicate lower intake	HEI-2010 / FoodAPS
Sodium	HEI moderation component measuring sodium intake	0–10 points; higher scores indicate lower intake	HEI-2010 / FoodAPS
Empty Calories	HEI moderation component measuring calories from solid fats, alcohol, and added sugars	0–20 points; higher scores indicate lower intake	HEI-2010 / FoodAPS
Rural	Household located in rural area	1 = rural; 0 = urban	FoodAPS

Variable	Definition	Coding / Measurement	Data Source
Married	Household reference person is married	1 = married; 0 = otherwise	FoodAPS
NH White	Household reference person is non-Hispanic White	1 = yes; 0 = otherwise	FoodAPS
NH Black	Household reference person is non-Hispanic Black	1 = yes; 0 = otherwise	FoodAPS
Hispanic	Household reference person is Hispanic	1 = yes; 0 = otherwise	FoodAPS
Household Education	Highest educational attainment in household	Ordinal category	FoodAPS
Income	Monthly household income	Continuous (US dollars)	FoodAPS
Children Age ≤ 5	Number of children aged five years or younger	Count	FoodAPS
Infants Age ≤ 1	Number of infants aged one year or younger	Count	FoodAPS
WIC-Eligible Woman	Presence of a WIC-eligible woman in household	1 = yes; 0 = no	FoodAPS
Nutrition Knowledge	Household nutrition knowledge score	Continuous index	FoodAPS
Primary Store Access	Primary food store located within 10 miles of residence	1 = yes; 0 = no	FoodAPS
Vehicle Ownership	Number of household vehicles available	Count	FoodAPS
Household Size	Total household members	Count	FoodAPS
Household Weight	FoodAPS household sampling weight	Continuous survey weight	FoodAPS

Note: HEI-2010 component scores follow the USDA Healthy Eating Index-2010 scoring system. For adequacy components, higher scores indicate greater consumption of recommended food groups. For moderation components

(Refined Grains, Sodium, and Empty Calories), higher scores indicate lower consumption and therefore better dietary quality. All variables are derived from the USDA National Household Food Acquisition and Purchase Survey (FoodAPS), 2012–2013.

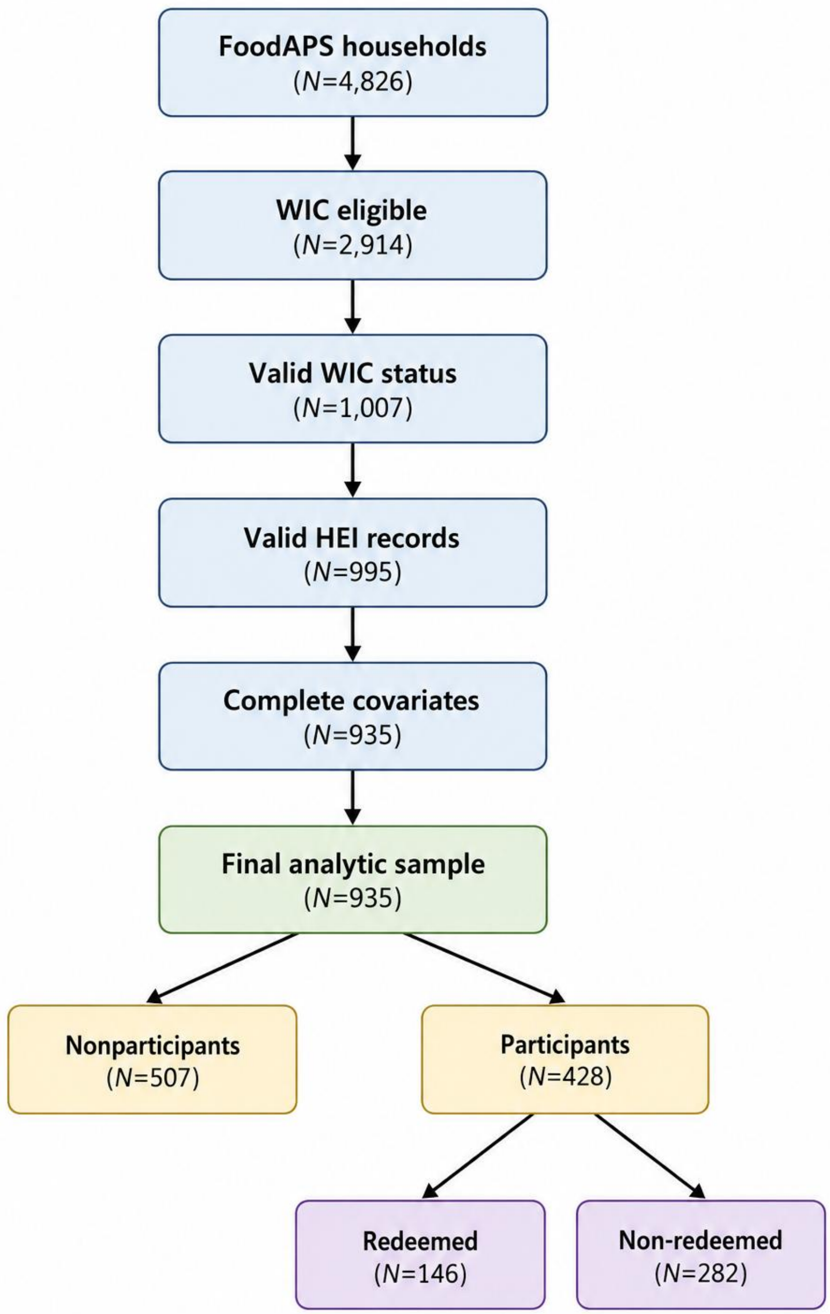

Figure A1. Sample Construction